

# AI-Augmented Demand Forecasting Engine Design (v3.29)

**Nature of this document:** Cross-domain methodology paper spanning Time-Series Analysis / Machine Learning / ERP / AI, describing erpilot v3.29's demand forecasting engine. This release closes the upstream gap in the v3.25.10–v3.28 planning chain by replacing customer-manual MPS entry with automatic forecasts from historical sales + LLM augmentation.

■ Prerequisites: [MRP\\_ALGORITHM\\_DESIGN\\_EN.md](#) (v3.25.10) / [THROUGHPUT\\_ACCOUNTING\\_DESIGN\\_EN.md](#) (v3.28)

## Abstract

erpilot's v3.25.10 → v3.28 planning chain assumed MPS is entered manually by the customer, undermining the objectivity of the entire IE/OR chain. This paper describes v3.29's demand forecasting engine, integrating Croston's (1972) classical intermittent demand algorithm, the Brown (1959) SES / Holt (1957) / Winters (1960) exponential smoothing family, and Hyndman & Koehler's (2006) MASE metric for automatic best-method selection. We also introduce Chatfield's (1989) regime change detection as an LLM-augmentation hook — when recent residuals exceed  $\pm 2\sigma$ , the system prompts customers to apply business judgment. Makridakis et al. (2018, 2020) M4/M5 competitions demonstrate that classical methods remain competitive with — sometimes beat — deep learning on most series, so this release excludes scipy/statsmodels/torch dependencies, implementing 5 methods in pure Python. 30 structural-invariant tests pass, including Hyndman-Koehler 2006's classic property  $\text{MASE} < 1 \Leftrightarrow \text{beats naive baseline}$ .

**Keywords:** Demand forecasting, Croston, Holt-Winters, MASE, intermittent demand, regime change, LLM augmentation

## 1. Introduction

### 1.1 The Fundamental Problem of Manual MPS

The 4 sprints v3.25.10 → v3.28 completed MRP-II, CLSP, and the TOC trilogy. But all plans start from an MPS that requires customers to fill in numbers like "produce 100 units of product A in W22." In practice:

- SMB owners lack data-analysis capability

- Sales history **exists**, but customers don't know how to use it
- Most parts have **intermittent demand** (zero for many weeks, then a spike) — intuitive entry is unreliable
- v3.27 explainability: "Why do we need so many M6 bolts next week?" always traces back to MPS — which the customer filled in, so erpilot can't ask 'are your numbers right?'

## 1.2 Why Not Deep Learning Directly

Makridakis (2018) M4 Competition results were striking: on a blind test of 100,000 time series:

| Method category                         | Average sMAPE | Rank     |
|-----------------------------------------|---------------|----------|
| Combine (weighted average of classical) | 11.4%         | 1st      |
| Exponential smoothing + ARIMA           | 11.9%         | 2nd      |
| Deep learning (LSTM, MLP)               | 13.7%         | bottom   |
| Naive                                   | 14.3%         | baseline |

Makridakis (2020) M5 using Walmart sales data: classical and ML perform similarly, but **classical is more robust, more interpretable, and cheaper to train**.

For SMB (short history, mostly intermittent), classical **wins more often**. We therefore choose:

-  Naive / Seasonal Naive baseline
-  SES / Holt / Holt-Winters
-  Croston (intermittent specialist)
-  No Prophet / N-BEATS / TFT for now (deferred to future sprints)

## 1.3 Why Croston Is Irreplaceable

Croston (1972, *J Op Res Soc* 23) founded the methodology for "intermittent demand forecasting." In ERP scenarios:

- Industrial screws: 0-0-0-50-0-0-30-0 per week
- Machine spare parts: once every 3 months
- Seasonal gifts: only November orders annually

SES **always over-forecasts** such data (low-bias error assumption fails). Croston decomposes:

$$\text{forecast} = (\text{smoothed\_demand\_when\_nonzero}) / (\text{smoothed\_interval\_between\_demands})$$

Syntetos & Boylan (2005) proved the original Croston has positive bias, requiring a  $(1 - \alpha/2)$  correction. We default to this corrected version.

## 2. Methodology

See [DEMAND\\_FORECASTING\\_DESIGN\\_ZH.md](#) §2 for full mathematical formulations of all 5 methods (Naive, SES, Holt, Holt-Winters, Croston) and accuracy metrics (MAPE, sMAPE, MASE per Hyndman-Koehler 2006).

Key design choices:

- **Auto-selection** by lowest MASE on holdout
- **Intermittent detection**: when `zero_fraction > 40%`, restrict candidates to {Naive, Croston} (per Syntetos-Boylan 2005)
- **Prediction intervals** via residual bootstrap,  $PI_{t+k} = \hat{y}_{t+k} \pm z_{\alpha} \sigma \sqrt{k}$
- **Regime change flag**:  $k$  of recent  $N$  residuals beyond  $\pm \theta \sigma$  (Chatfield 1989 §13)

## 3. Implementation

```
backend/app/services/demand_forecasting.py    (~600 lines, pure Python)
├─ ForecastMethod (Enum: NAIVE / SEASONAL_NAIVE / SES / HOLT / HOLT_WINTERS / CROSTON / AUTO)
├─ ForecastResult / BacktestResult / RegimeChangeAlert (dataclasses)
├─ 5 forecasting algorithms (each ~30-80 lines)
├─ 3 accuracy metrics (MAPE, sMAPE, MASE)
├─ backtest / auto_select_method
├─ prediction_intervals_residual
├─ detect_regime_change (LLM augmentation hook)
└─ forecast (top-level API)
```

## 4. Validation

### 4.1 30 Structural Invariant Tests (10 categories, all pass)

| Category                | Tests | Coverage                                                              |
|-------------------------|-------|-----------------------------------------------------------------------|
| 1. Naive baseline       | 4     | Flat output / empty / seasonal rotation / fallback                    |
| 2. SES (Brown 1959)     | 3     | $\alpha=1$ tracking / $\alpha\approx 0$ smoothing / constant recovery |
| 3. Holt (1957)          | 2     | Linear extrapolation / horizon linearity                              |
| 4. Holt-Winters (1960)  | 2     | Perfect cycle recovery / short-history fallback                       |
| 5. Croston (1972)       | 4     | All zero / rate recovery / flat output / single event                 |
| 6. Accuracy metrics     | 6     | MAPE / sMAPE / MASE classic properties                                |
| 7. Auto-selection       | 2     | Intermittent restriction + lowest MASE wins                           |
| 8. Prediction intervals | 2     | Widens with horizon / zero residual zero width                        |
| 9. Regime change        | 2     | Detect spike / no false positive on stable                            |
| 10. API integration     | 3     | Full ForecastResult / horizon=0 / explicit method                     |

### 4.2 Results

30/30 tests pass. Sprint cumulative: 442/442 smoke tests pass.

## 5. Limitations and Future Work

| Topic                                                 | Why not                                          | Future direction                                     |
|-------------------------------------------------------|--------------------------------------------------|------------------------------------------------------|
| Modern ML (Prophet / N-BEATS / TFT)                   | M4 shows classical still competitive; heavy deps | v3.30+: optional deps, GPU env only                  |
| Time-series CV                                        | Holdout only, not rolling                        | Hyndman 2021 §5.4 rolling CV                         |
| Hierarchical reconciliation                           | product/category/total inconsistency             | Wickramasuriya, Athanasopoulos & Hyndman (2019) MinT |
| Full auto-tuning of ( $\alpha$ , $\beta$ , $\gamma$ ) | Only $\alpha$ grid search                        | Nelder-Mead via scipy                                |
| Multiplicative seasonality                            | Additive only                                    | Holt-Winters multiplicative variant                  |
| External regressors                                   | Univariate only                                  | ARIMAX                                               |
| ETS state-space framework                             | Not integrated                                   | Hyndman, Koehler, Snyder & Grose (2002)              |
| Actual LLM augmentation                               | Spec defined, API not wired                      | v3.30: prompt template + ConfirmCard flow            |

### Edge Cases

- History < 2 seasons: HW auto-falls-back to Holt
- All-zero history: all methods return 0
- Single non-zero event: Croston uses mean / expected interval
- Negative demand (returns): not specially handled; recommend "net sales" input

## 6. Legal Notice / 法律聲明

### Important: Legal Nature of Statistical Forecasts

This module produces statistical demand forecasts. All forecasts carry inherent uncertainty and are NOT:

- Guarantees of future demand:** point forecasts are algorithm-computed expected values based on historical data; actual demand may differ significantly. **Prediction intervals** are

probabilistic approximations — a 95% PI does **NOT** guarantee 95% capture rate (especially when data-generating process changes)

2. **Basis for procurement commitments:** customers **must NOT** use this module's forecasts as the **sole basis** for supplier commitments, employee overtime, or customer delivery guarantees. Major decisions must integrate market conditions, specific customer SOs, and business intelligence
3. **Evidence in supply-chain disputes:** this module's output **must NOT** be cited as "the forecast said X, so the supplier should have..." in disputes. Forecasts are fundamentally **estimates**, not contractual commitments
4. **Prediction of unforeseeable events:** no statistical method can predict:
  - Natural disasters (earthquakes, floods, typhoons)
  - Policy changes (tariffs, regulations, trade wars)
  - Customer business changes (bankruptcy, mergers, relocation)
  - Competitive landscape shifts (new entrants, technology breakthroughs)
  - Black Swan events (Taleb 2007)

## 1. Algorithm Limitations

- **Stationarity assumption:** all methods assume the data-generating process is stable within the forecast horizon. **Regime change** detection is a rough indicator, NOT a replacement for business judgment
- **MASE < 1 doesn't guarantee correct decisions:** MASE is "relative to naive baseline"; MASE 0.5 may still be insufficient in absolute accuracy for major decisions
- **Mis-calibrated PIs:** when residuals exhibit autocorrelation or heteroscedasticity, PIs are inaccurate
- **Auto-selection overfitting:** choosing methods by holdout MASE may be "lucky" on holdout but unstable in production

## 2. LLM Augmentation Warning (when enabled in v3.30+)

When `detect_regime_change` triggers LLM review:

- LLM may **hallucinate** business reasons (fabricate customer/competition scenarios)
- LLM's qualitative adjustments should be **advisory only**, never auto-overwriting forecasts
- All LLM suggestions must go through **ConfirmCard** (human confirmation)
- LLM-stated reasons (e.g., "Customer X cancelled") **do NOT constitute factual statements about Customer X** — avoid defamation risk

## 3. Croston / Intermittent Demand Special Notice

Croston (1972) assumes **non-zero demand events** are a random process. In practice:

- If demand events have **structural causes** (e.g., end-of-month inventory top-up), statistical methods may still fail
- Syntetos-Boylan (2005) bias correction is correct in mean but still biased in quantile estimation
- For **extremely sparse data** (e.g., once-per-year), no statistical method is reliable — use business intelligence

## 4. Responsibility for MPS Integration

This module can auto-generate MPS suggestions. But:

- **MPS final authority should rest with the production controller**, not be auto-generated
- Workflow must follow **ConfirmCard**: forecast → MPS suggestion → controller review → confirm → flow into MRP
- **Major decisions** (annual bulk material procurement, capacity investment) require additional review

## 5. Disclaimer Clause

To the maximum extent permitted by applicable law, erpilot assumes no liability for:

- **Over-purchasing / line stoppage losses** from acting on these forecasts
- **Customer delivery defaults / penalties** from forecast inaccuracy
- **Erroneous business judgments** influenced by LLM suggestions
- **Planning failures** from undetected regime changes
- **Any consequences** from using auto-generated MPS without controller review

## 6. Cumulative Applicability of Predecessors

This version completes the upstream gap of the v3.25.10 → v3.28 planning chain. **All disclaimers** in predecessor §6 documents **apply cumulatively** here.

## Recommended Practice

- Treat forecasts as "**starting suggestions**" not "final MPS"
- Compare actual vs forecast monthly; update method selection and parameters
- High-value / high-risk parts (Class A) must have human review
- Demand outside PI → proactively contact customer to verify cause
- Retain forecast-accuracy history as reference for negotiation but **not as legal evidence**

## 7. References

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## 8. Changelog

| Version       | Date       | Change                                                                            |
|---------------|------------|-----------------------------------------------------------------------------------|
| v3.25.10      | 2026-05-20 | MRP-II (based on manual MPS)                                                      |
| v3.26 - v3.28 | 2026-05-20 | CLSP / TOC trilogy (still on manual MPS)                                          |
| v3.29         | 2026-05-20 | <b>This release:</b> automatic demand forecasting engine, closes MPS upstream gap |
| Future v3.30+ | TBD        | LLM augmentation integration / Prophet evaluation / Hierarchical                  |
| Future v3.31+ | TBD        | Rolling CV / ETS state-space / Multiplicative seasonality                         |

**Last updated:** 2026-05-20 (v3.29) **Authors:** erpilot engineering team (with TS analysis / ML / ERP / AI cross-domain academic methodology)      **Version:** 1.0      **Chinese version:** [DEMAND\\_FORECASTING\\_DESIGN\\_ZH.md](#)